TPN analysis: successor computation - conditioning

- Condition D to the fact that t_0 fires first:

Questi rappresentano i vincoli che imponiamo a D per l'attivazione di t0.

Questo è come quello della slides sopra

$$D_{t_0} = \begin{cases} D \\ \tau(t_0) - \tau(t_h) \leq 0 \\ \forall\, t_h \in T_{\mathrm{e}}(m) \setminus \{t_0\} \end{cases} = \begin{cases} \tau(t_i) - \tau(t_j) \leq b_{ij} \\ \tau(t_0) - \tau(t_h) \leq 0 \\ \forall\, t_i, t_j \in T_{\mathrm{e}}(m) \cup \{t_\star\} \mid t_i \neq t_j \\ \forall\, t_h \in T_{\mathrm{e}}(m) \setminus \{t_0\} \end{cases}$$

$$= \begin{cases} \tau(t_i) - \tau(t_j) \leq b_{ij} \\ \tau(t_0) - \tau(t_h) \leq \min\{0, b_{0h}\} \\ \forall\, t_i, t_j \in T_{\mathrm{e}}(m) \cup \{t_\star\} \mid t_i \neq t_j,\ \forall\, t_h \in T_{\mathrm{e}}(m) \setminus \{t_0\} \end{cases}$$

- Represent D_{t_0} in normal form and let B_{ij} be its coefficients (quadratic complexity wrt the number of enabled transitions):

$$D_{t_0} = \begin{cases} \tau(t_i) - \tau(t_j) \leq B_{ij} \\ \tau(t_0) - \tau(t_i) \leq B_{0i} \\ \forall\, t_i, t_j \in T_{\mathrm{e}}(m) \cup \{t_\star\} \mid t_i \neq t_j,\ \ t_i \neq t_0 \end{cases}$$

TPN analysis: successor computation - time advancement

- Reduce times to fire of persistent transitions in S' by the time elapsed in S, i.e., $\tau'(t_i) = \tau(t_i) - \tau(t_0)\ \forall\, t_i \in T_{\mathrm{e}}(m) \cap T_{\mathrm{p}}(S') \setminus \{t_0\}$ (where $T_{\mathrm{p}}(S')$ is the set of transitions that are persistent in S'):

=TRASCORSO

In questo passaggio si sono fatte le sostituzioni con quello sopra

$$D_{t_0} = \begin{cases} \tau(t_i) - \tau(t_j) \leq B_{ij} \\ \tau(t_i) - \tau(t_0) \leq B_{i0} \\ \tau(t_0) - \tau(t_i) \leq B_{0i} \\ \tau(t_i) - \tau(t_\star) \leq B_{i\star} \\ \tau(t_\star) - \tau(t_i) \leq B_{\star i} \\ \tau(t_0) - \tau(t_\star) \leq B_{0\star} \\ \tau(t_\star) - \tau(t_0) \leq B_{\star 0} \\ \forall\, t_i, t_j \in T_{\mathrm{e}}(m) \cap T_{\mathrm{p}}(S') \cup \{t_\star\} \mid \\ t_i \neq t_j,\ t_i \neq t_0,\ t_i \neq t_\star \end{cases}$$

Timer diversi rispetto a quello che spara

Timer che spara rispetto agli altri

Timer diversi da quello che spara rispetto al ground

Timer che spara rispetto al ground

$$D'_{t_0} = \begin{cases} \tau'(t_i) - \tau'(t_j) \leq B_{ij} \\ \tau'(t_i) \leq B_{i0} \\ -\tau'(t_i) \leq B_{0i} \\ \tau'(t_i) + \tau(t_0) \leq B_{i\star} \\ -\tau(t_0) - \tau'(t_i) \leq B_{\star i} \\ \tau(t_0) - \tau(t_i) \leq B_{0i} \\ \tau(t_0) - \tau(t_\star) \leq B_{0\star} \\ \tau(t_\star) - \tau(t_0) \leq B_{\star 0} \\ \forall\, t_i, t_j \in T_{\mathrm{e}}(m) \cap T_{\mathrm{p}}(S') \cup \{t_\star\} \mid \\ t_i \neq t_j,\ t_i \neq t_0,\ t_j \neq t_\star \end{cases}$$

La loro differenza non è cambiata perché sono transizioni persistenti

tau(t0) diventa il nuovo ground, cioé il tempo di ingresso nello stato.

Sostituzione con i nuovi valori sopra

TPN analysis: successor computation - projection

- Eliminate the time to fire of the fired transition through a projection:

$$D''_{t_0} = \begin{cases} \tau'(t_i) - \tau'(t_j) \leq B_{ij} \\ \tau'(t_i) \leq B_{i0} \\ -\tau'(t_i) \leq B_{0i} \\ \forall\, t_i, t_j \in T_{\mathrm{e}}(m) \cup \{t_\star\} \mid t_i \neq t_j,\ \ t_i \neq t_0 \end{cases}$$

- Let C_{ij} be the normal form coefficients (i.e., $C_{ij} = B_{ij}$, $C_{i\star} = B_{i0}$, $C_{\star i} = B_{0i}$):

Come avevamo detto t0 diventa il gound

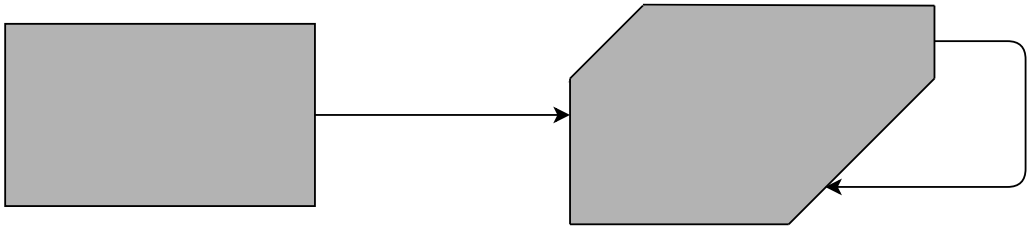
$$D''_{t_0} = \begin{cases} \tau'(t_i) - \tau'(t_j) \leq C_{ij} \\ \tau'(t_i) - \tau'(t_\star) \leq C_{i\star} \\ \tau'(t_\star) - \tau'(t_i) \leq C_{\star i} \\ \forall\, t_i, t_j \in T_{\mathrm{e}}(m) \cap T_{\mathrm{p}}(S') \cup \{t_\star\} \mid t_i \neq t_j,\ \ t_i \neq t_\star \end{cases}$$

- Times-to-fire of transitions not enabled by m' can be similarly eliminated

TPN analysis: successor computation - newly enabling

- Add the time to fire of each newly-enabled transition:

$$D' = \begin{cases} D''_{t_0} \\ \tau'(t_j) - \tau'(t_\star) \leq LFT(t_j) \\ \tau'(t_\star) - \tau'(t_j) \leq -EFT(t_j) \\ \forall\, t_j \in T_{\mathrm{n}}(S') \end{cases}$$

  where $T_{\mathrm{n}}(S')$ is the set of transitions newly-enabled in S'

- The firing domain D' of the successor class S' is still in DBM form! $\Rightarrow$
  $\Rightarrow$ The DBM form is closed wrt the steps of successor computation!
  $\Rightarrow$ The reachability relation among state classes can be enumerated!